Contractions, punctuation & typos

02 **Match an intent**

Synonyms & phrase scoring

03 **Return a response**

Local date/time or safe fallback

TRY A PROMPT

? What can you do?

◷ Tell me the time

✳ Tell me a joke

Type a message...

Send ↑

ENTER TO SEND

RULE-BASED · NO GENERATIVE AI

Start with a clear, testable brief

[input → intent → response]

// The Objective

Build a transparent AI chatbot that relies on explicit rules rather than generative models. The goal was to master the "foundations of AI": input sanitization, intent classification, and deterministic state management.

Every response must be explainable, repeatable, and delivered through a continuous conversational loop.

// Scope & Requirements

> **Eight Intent Groups**
Greetings, help, identity, time, date, joke, thanks, and farewell.

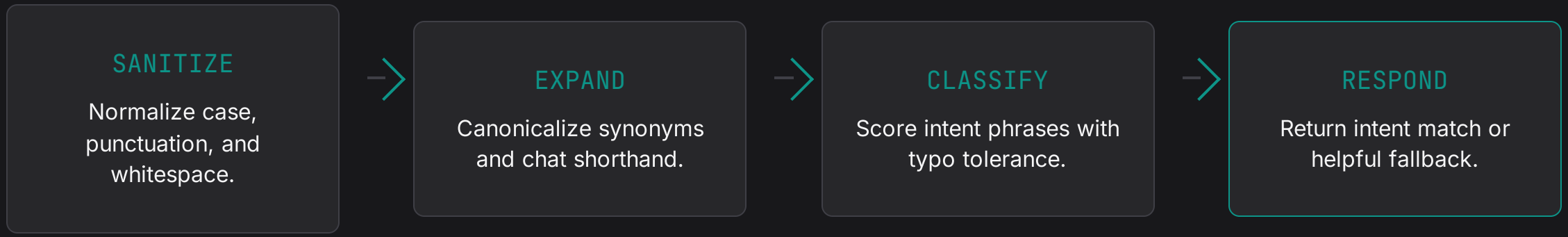
> **Continuous Lifecycle**
Looping chat interface with clean exit commands (exit, quit, bye).

> **Input Discipline**
Normalization of case, punctuation, and whitespace before matching.

> **Safe Fallbacks**
Graceful handling of empty or unknown inputs without crashing.

A transparent pipeline, end to end

[input → intent → response]



From CLI to a browser experience

[input → intent → response]

// Product decisions

Two ways to run it

A standard-library Python CLI for terminal use, plus a responsive static web app.

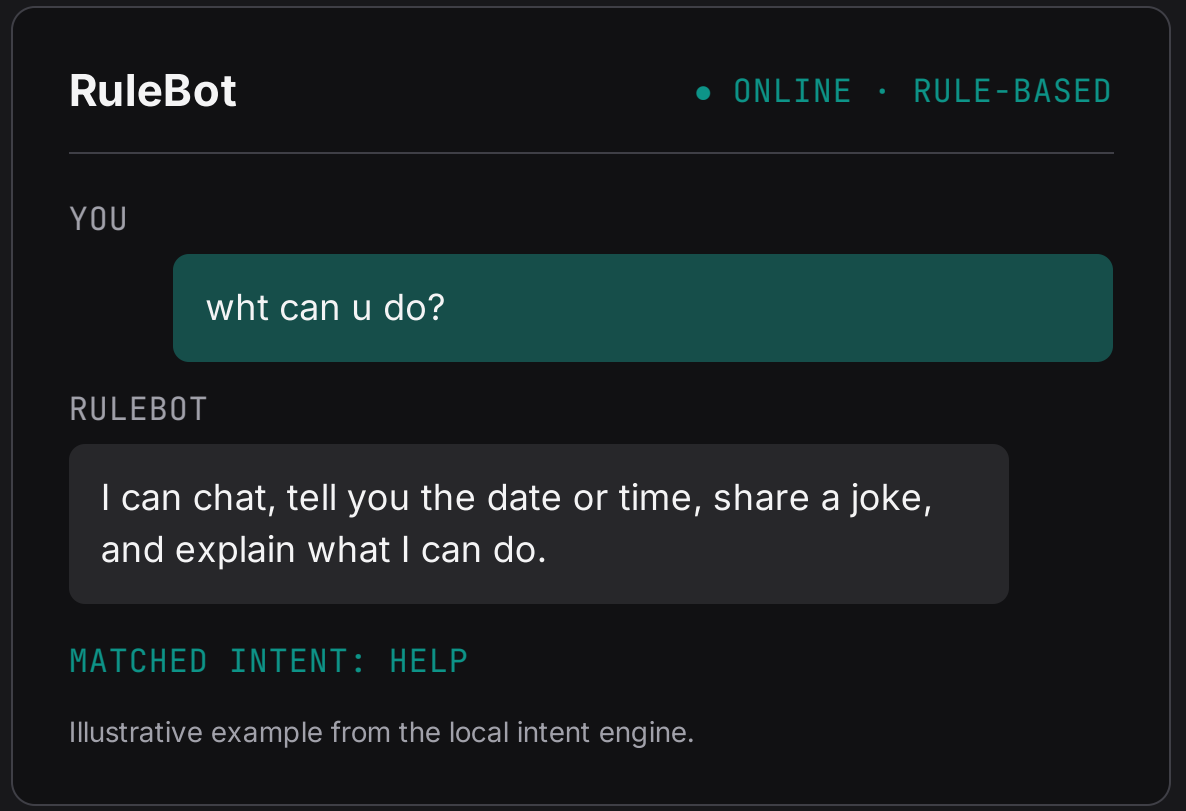
Built for repeated turns

Quick prompts make supported intents discoverable; new chat restarts the conversation after a clean exit.

Client-side by design

The browser app can run on static hosting without a chatbot backend or server-side conversation store.

// Example interaction



Illustrative example from the local intent engine.

DEPLOYED ON RENDER · LIVE WEB APP

More tolerant language—without an API

[input → intent → response]

RuleBot upgrades basic exact-string matching with a local NLP layer. It uses Levenshtein distance for typo tolerance and token mapping for common chat shorthand.

User Input	Normalization & Synonyms	Scored Intent
"wht's today's date?"	what is today's date	**date (score: 4)**
"thx, crack a joke"	thanks, crack a joke	**joke (score: 6)**
"helo, tel time"	hello, tell time	**time (score: 4)**
"hiya! whos rulebot?"	hello who is rulebot	**identity (score: 4)**

Built to be explainable and testable

[input → intent → response]

01 / EXPLICIT RULES

Lexical Knowledge Base

A dictionary-driven intent map stores eight distinct categories. Every response is tied to an auditable rule rather than a black-box weights file.

02 / DYNAMIC STATE

Injectable Real-Time

Date and time handlers use an injectable clock, allowing precise regression testing of time-dependent responses without waiting for reality.

03 / RELIABILITY

Regression Suites

14 Python unit tests and 8 browser-side Node.js tests cover normalization, exit triggers, fallback safety, and intent precedence.

04 / CONTROL FLOW

Safe Exits

Exit commands are recognized as high-priority full commands. Both environments provide a graceful teardown and session restart.

Ship simply. Learn deliberately.

[input → intent → response]

// Project Architecture

Private GitHub Repo	Source
Render Static Site	Host
Custom NLP Lib	Logic
Automated CI/CD	Pipeline

RuleBot demonstrates that a production-ready project is about the whole lifecycle: from clean code and local testing to secure hosting and responsive design.

Project Status

✓

All requirements met.
Tests passing.
Deployed.

```
return 0;
```